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Guru Nanak Institute of Technology
An Autonomous Institute under MAKAUT
2024
EVEN SEMESTER 2023-2024
CH(EC)201 - ENGINEERING CHEMISTRY

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) For which of the following Bohr's Theory is applicable?

(1) CO1 BL2

- a) H
- b) He⁺
- c) Li²⁺
- d) all of the above

1.ii) Rusting of Iron is:

(1) CO4 BL1

- a) Dry corrosion
- b) wet corrosion
- c) protective corrosion
- d) non-protective corrosion

1.iii) Example of natural polymer is:

(1) CO2 BL1

- a) PVC
- b) Nylon
- c) Cellulose
- d) bakelite

1.iv) Atomic size of the elements in the periodic table:

(1) CO1 BL2

- a) decreases from left to right
- b) increases from left to right
- c) decreases down the group
- d) does not change down the group

1.v) Maximum work can be obtained in an isothermal process when it is:

(1) CO4 BL1

- a) Irreversible
- b) adiabatic
- c) reversible
- d) cyclic

- 1.vi) Absorbance of a sample is calculated from: (1) CO5 BL1
 a) Raoult's Law
 b) Lambert-Beer's Law
 c) Pauli's Exclusion Principle
 d) Golden rule

- 1.vii) A process in which pressure remains constant is called (1) CO1 BL1
 a) Isochoric process
 b) Isobaric process
 c) Isothermal process
 d) Adiabatic process

- 1.viii) Bakelite is (1) CO3 BL1
 a) Addition polymer
 b) Condensation polymer
 c) Thermoplastic polymer
 d) Thermosetting polymer

- 1.ix) The possible set of quantum no. for the unpaired electron of sodium is (1) CO1 BL2
 a) $n=2, l=1, m=0$
 b) $n=2, l=1, m=1$
 c) $n=3, l=1, m=1$
 d) $n=3, l=0, m=0$

- 1.x) Thermodynamically boiling point is (1) CO2 BL2
 a) Intensive property
 b) Extensive property
 c) both intensive and extensive
 d) None of the above

- 1.xi) Free energy change for a spontaneous chemical process is (1) CO2 BL2
 a) Zero
 b) negative
 c) positive
 d) unchanged

- 1.xii) Racemization occurs in which of the following? (1) CO5 BL2
 a) E1 reaction
 b) SN2 reaction
 c) SN1 reaction
 d) free radical halogenation

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. Answer the following (5)
 a) Though F is higher electronegative than Cl, but Cl has higher electron affinity than F. Why? (3) CO1 BL2

- b) Arrange the following ions in increasing order of their size: P^{3-} , Ca^{2+} , S^{2-} , Cl^- , K^+ . (2) CO1 BL2

3. Answer the following (5)

- a) What are the basic differences between n-type and p-type semiconductors? (3) CO4 BL1

- b) What is F-centre? How does it affect crystal properties? (2) CO4 BL1

4. Distinguish between SN1 and SN2 reactions. (5) CO3 BL2

5. Write down the Nernst equation and calculate the EMF of a Daniel cell when the concentration of $ZnSO_4$ and $CuSO_4$ are 0.01 M and 0.1 M respectively. The standard electrode potential of the cell is 1.1 V. (5) CO3 BL3

6. Define Ionization enthalpy. How do you explain its variation in a period and in a group? (5) CO2 BL2

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

7. Answer the following (15)

- a) Formic acid is stronger acid than acetic acid. Explain (3) CO1 BL2

- b) Show the general graphic representations of a UV-Vis and IR spectra with proper axes description. (2) CO5 BL1

- c) State Markovnikov rule. What is peroxide effect? (3) CO1 BL2

- d) What is conducting polymer? Give example. (2) CO2 BL1

- e) What is enthalpy? What is its significance? How it is related to the internal energy? (5) CO4 BL1

8. Answer the following (15)

- a) Cationic Frenkel defects are more common than anionic Frenkel defects. Explain with example (3) CO4 BL1

- b) Find out ΔG for the cell reaction occurring in the cell, $Zn|ZnSO_4 (0.1M)||AgNO_3 (0.1M)|Ag$ at 25 °C. Given $E^\circ_{cell} = 1.56 V$. (2) CO1 BL3

- c) What is inductive effect? How does it influence the polarity of a molecule? (3) CO1 BL2

- d) IE1 for Mg > IE1 for Na, but IE2 for Na > IE2 for Mg. Explain (2) CO1 BL2

- e) What is hard water? What are the types of hardness? How it can be removed? (4) CO3 BL1

- f) What is the common reagent in preparation of Paracetamol and Aspirin? (1) CO1 BL1

9. Answer the following (15)

- a) Discuss the working principle of Acid storage cell with detail reactions for discharging and recharging cycle (5) CO4 BL2
- b) What are the differences between elastomer and fibre? (3) CO2 BL1
- c) Draw the polymeric chain of Nylon 6,6. (2) CO2 BL2
- d) Inversion of configuration occurs with S_N2 reaction and racemization occurs with S_N1 reaction. Explain with mechanism. (5) CO1 BL3
10. Answer the following (15)
- a) State 2nd law of Thermodynamics. The efficiency of a reversible heat engine depends only on the temperature of source and sink. Explain (5) CO4 BL2
- b) Anodic protection is a better method than cathodic protection. Justify. (2) CO4 BL1
- c) 250 ml of sample water contains 12.5 mg of $CaSO_4$ and 15 mg of $Mg(HCO_3)_2$. What will be temporary and permanent hardness? Find out the total hardness also. (3) CO3 BL3
- d) Distinguish between paramagnetism and ferromagnetism with suitable example. (3) CO1 BL1
- e) What is internal energy? (2) CO4 BL1
11. Write short notes on any 3 of the following: (15) CO5 BL4
- a) Spectroscopic analysis of materials
- b) 1st law of thermodynamics
- c) Biodegradable polymers
- d) Aufbau Principle and Hund's rule
- e) Corrosion control



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Guru Nanak Institute of Technology
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2024

EVEN SEMESTER 2023-2024
EC201 - CIRCUIT THEORY AND NETWORK

TIME ALLOTTED: 3Hours

FULL MARKS: 70

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A**(Multiple Choice Type Questions)**Answer any ten from the following, choosing the correct alternative of each question: $10 \times 1 = 10$

(1) CO2 BL3

1.i) The internal resistance of an ideal voltage source is

- a) 0
- b) infinite
- c) 1
- d) 0.5

(1) CO2 BL2

1.ii) Thevenin's theorem can be applied to calculate the current in

- a) a linear load only
- b) any load
- c) a passive load
- d) a bilateral load only

(1) CO4 BL2

1.iii) The s-domain equivalent of the inductor reduces to an inductor with impedance

- a) sL
- b) L
- c) s^2L
- d) s^3L

(1) CO4 BL5

1.iv) The Laplace transform of $\sin(\omega t)$ is:

- a) $\sin(\omega s)$
- b) $\cos(\omega s)$
- c) $\omega / (s^2 + \omega^2)$
- d) $s / (s^2 + \omega^2)$

(1) CO1 BL2

1.v) In a series resonant R-L-C circuit, the power factor at resonance is

- a) 1
- b) 0.5
- c) 0
- d) infinity

1. vi) The hybrid parameter h_{22} in terms of Z-parameters is given by: (1) CO5 BL6

- a) $1/Z_{22}$
- b) $-Z_{11}/Z_{22}$
- c) Z_{12}/Z_{22}
- d) none of these

1. vii) A graph of an electrical network has 4 nodes and 7 branches. The number of links l , with respect to the chosen tree would be (1) CO3 BL2

- a) 4
- b) 3
- c) 2
- d) 5

1. viii) As $X_L = X_C$ in a series resonance circuit, the impedance is _____ (1) CO1 BL2

- a) purely resistive
- b) purely capacitive
- c) purely inductive
- d) capacitive and inductive

1. ix) The mesh analysis is based on

- a) KCL and KVL
- b) KVL
- c) KCL
- d) Ohm's law and KVL

1. x) Time constant of RC circuit is

- a) RC
- b) R/C
- c) C/R
- d) 1/RC

1. xi) According to the graph theory of loop analysis, how many equilibrium equations are required at a minimum level in terms of number of branches (b) and number of nodes (n) in the graph? (1) CO3 BL2

- a) $b-(n-1)$
- b) $b+(n-1)$
- c) $b^*(n-1)$
- d) $b-(n+1)$

1. xii) Maximum power transfer occurs at circuit efficiency of (1) CO2 BL3

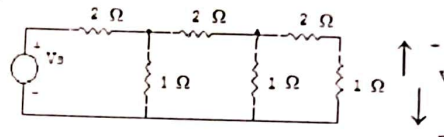
- a) 100 %
- b) 50 %
- c) 25 %
- d) 75 %

Group B
(Short Answer Type Questions)

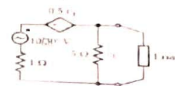
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2. Find out V_s in terms of V . (Answer any three of the following) 3x5=15

(5) CO2 BL3



3. Find the Thevenin's equivalent circuit of the network shown in figure below across the load. (5) CO2 BL3

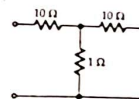


4. What is the resonance frequency of a series RLC circuit where $R=10 \Omega$, $L=25 \text{ mH}$, $C=100 \mu\text{F}$? Evaluate the Q factor also. (5) CO1 BL3

5. With reference to figure below, draw the graph and write down the tie-set matrix. (5) CO3 BL4



6. Two identical T sections, as one shown in figure below are connected in cascade. Determine the Z-parameters of the combination. (5) CO5 BL4



Group C
(Long Answer Type Questions)

7. Describe about the transient response in series R-L circuit having DC excitation (charging and discharging both with no initial condition). (15) CO3 BL4

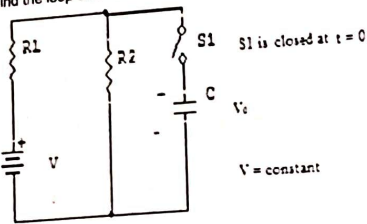
8a. What do you mean by Laplace Transform ? (3) CO4 BL5

8b. Obtain the Laplace transform of $[e^{\theta t} \sin(\omega t)]$ (where $\theta =$ constant) (4) CO4 BL5

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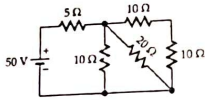
8c. Find the loop currents in S domain and also in matrix form.

(8) CO4 BL5



9a. Show the validity of reciprocity theorem in the network shown in figure below.

(9) CO2 BL3

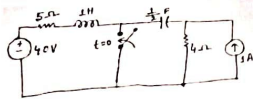


9b. Derive the expression for open circuit impedance parameters (Z-parameters) in two port network.

(6) CO5 BL3

10a. Find the current $i(t)$ for the circuit shown in Fig. Assume that the circuit has reached steady state at $t = 0^-$.

(8) CO4 BL3



10b. A capacitor of 10 pF is connected to a voltage source of 100 V. If the distance between the capacitor plates is reduced to 50%, while it remains connected to the 100 V supply, the value of potential gradient in the second case will be

(3) CO2 BL4

10c. In the given circuit, the two-port network has the impedance matrix

(4) CO3 BL3

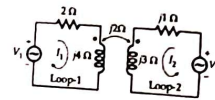
$$[Z] = \begin{bmatrix} 40 & 60 \\ 60 & 120 \end{bmatrix}$$

Calculate the value of Z_L for maximum power transfer.

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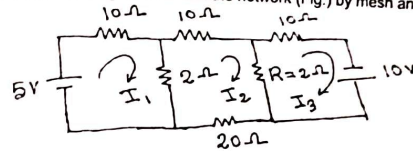
11a. Find V_2 in the circuit of figure below such that current in the left hand loop (loop-1) is zero. Assume $V_1 = 5\angle 0^\circ$ V.

(7) CO4 BL1



11b. Find the voltage across R in the network (Fig.) by mesh analysis.

(4) CO1 BL3



11c. Explain about Self and Mutual inductance.

(4) CO5 BL6

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Roll Number

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2024

EVEN SEMESTER 2023-2024

M(EC)201 - ENGINEERING MATHEMATICS - II

TIME ALLOTTED: 3Hours

FULL MARKS: 70

*Instructions to the candidate:**Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Group A****(Multiple Choice Type Questions)**

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) (1) CO2 BL1

Order and degree of $\frac{d^2y}{dx^2} = \left\{ y^4 + \left(\frac{dy}{dx} \right)^3 \right\}^{\frac{1}{2}}$ is

- a) 3,2
- b) 1,2
- c) 2,3
- d) 2,2

1.ii) I.F of $\frac{dy}{dx} - y = \sin x$ is (1) CO3 BL2

- a) e^{3x}
- b) e^{2x}
- c) e^x
- d) e^{-x}

1.iii) C.F of $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} = 3x$ is (1) CO3 BL1

- a) $Ae^x + Be^{3x}$
- b) $Ae^{-x} + Be^{3x}$
- c) $Ax + Bx^2$
- d) None of these

1.iv) The particular integral of $\frac{d^2y}{dx^2} + y = \cos x$ is

(1) CO1 BL1

- (a) $-\sin x$
 (b) $-\cos x$
 (c) $-\frac{1}{2} \sin x$
 (d) $\frac{1}{2} \sin x$

1.v) $L\left\{\frac{e^{-ax}}{a}\right\} = ?$

(1) CO2 BL2

- a) $\frac{1}{a(p+a)}$
 b) $\frac{1}{p-a}$
 c) $\frac{1}{a(p-a)}$

1.vi) Which of the following is true?

(1) CO1 BL1

- a) $L^{-1}\{f(s-a)\} = e^{-at}L^{-1}\{f(s)\}$
 b) $L^{-1}\{f(s-a)\} = L^{-1}\{f(s)\}$
 c) $L^{-1}\{f(s-a)\} = e^{at}L^{-1}\{f(s)\}$
 d) $L^{-1}\{f(s+a)\} = e^{at}L^{-1}\{f(s)\}$

1.vii) $L(e^{-2t} \cos t)$ is equal to

(1) CO2 BL2

- a) $\frac{s+2}{s^2+4s+5}$
 b) $\frac{s}{s^2+4s+5}$
 c) $\frac{s+1}{s^2+4s-1}$
 d) $\frac{s+3}{s^2+4s-1}$

1.viii) The equation $\frac{dy}{dx} + Py = Q$ is a linear equation of first order if

(1) CO1 BL1

- (a) P is a function of x and Q is a function of y
 (b) P is a function of y and Q is a function of x
 (c) P and Q are both functions of x
 (d) None of these

1.ix) The substitution $y = e^x$ transforms the differential equation $y^2 \frac{dy}{dx} - 5y = \ln x$ to

(1) CO1 BL1

- (a) $\frac{d^2y}{dz^2} - \frac{dy}{dz} - 5y = z$, (b) $\frac{d^2y}{dz^2} - \frac{dy}{dz} + 5y = z$
 (c) $\frac{d^2y}{dz^2} - \frac{dy}{dz} + 5y = 0$, (d) $\frac{d^2y}{dz^2} - \frac{dy}{dz} - 5y = z$

1.x) Rounding off the number 0.03709157 correct upto 5 significant figure is

(1) CO2 BL2

- a) 0.03709
 b) 0.037091
 c) 0.037092
 d) 0.0370

1.xi) If $y(x) = x^2$ and h is a step length, then find which one of the following is true

(1) CO1 BL1

- a) $\Delta f(x) = h(x+2h)$
 b) $\Delta f(x) = 2h(x+h)$
 c) $\Delta f(x) = h(x+h)$
 d) $\Delta f(x) = h(2x+h)$

1.xii) Trapezoidal rule can be applied if the number of equal sub-intervals of the interval of integration is

(1) CO3 BL3

- a) odd
 b) even
 c) any
 d) none

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. Solve $y' = 2px + yp^2$

(5) CO3 BL2

3. Apply the method of variation of parameters to solve

(5) CO3 BL3

$$\frac{d^2y}{dx^2} + 4y = \tan 2x$$

4. Using Laplace transform evaluate $\int_0^\infty t^2 e^{-st} \sin t \, dt$

(5) CO3 BL3

5. Find the missing term in the following table

(5) CO2 BL2

x	0	1	2	3	4
f(x)	1	3	9	-	81

1.iv) The particular integral of $\frac{d^2y}{dx^2} + y = \cos x$ is

- (a) $\frac{1}{2}\sin x$
 (b) $\frac{1}{2}\cos x$
 (c) $\frac{1}{2}x\sin x$
 (d) $x\sin x$

(1) CO1 BL1

1.v) $L\left\{\frac{e^{-ax}}{a}\right\} = ?$

- a) $\frac{1}{a(p+a)}$
 b) $\frac{1}{p-a}$
 c) $\frac{1}{a(p-a)}$

(1) CO2 BL2

1.vi) Which of the following is true?

- a) $L^{-1}\{f(s-a)\} = e^{-at}L^{-1}\{f(s)\}$
 b) $L^{-1}\{f(s-a)\} = L^{-1}\{f(s)\}$
 c) $L^{-1}\{f(s-a)\} = e^{at}L^{-1}\{f(s)\}$
 d) $L^{-1}\{f(s+a)\} = e^{at}L^{-1}\{f(s)\}$

(1) CO1 BL1

1.vii) $L(e^{-2t} \cos t)$ is equal to

- a) $\frac{s+2}{s^2+4s+5}$
 b) $\frac{s}{s^2+4s+5}$
 c) $\frac{s+1}{s^2+4s-1}$
 d) $\frac{s+3}{s^2+4s-1}$

(1) CO2 BL2

1.viii) The equation $\frac{dy}{dx} + Py = Q$ is a linear equation of first order if

- (a) P is a function of x and Q is a function of y
 (b) P is a function of y and Q is a function of x
 (c) P and Q are both functions of x
 (d) None of these

(1) CO1 BL1

1.ix) The substitution $x = e^t$ transforms the differential

$$\text{equation } x^2 \frac{d^2y}{dx^2} - 5x = \ln x \text{ to}$$

- (a) $\frac{d^2y}{dz^2} + \frac{dy}{dz} - 5y = z$, (b) $\frac{d^2y}{dz^2} - \frac{dy}{dz} + 5y = z$,
 (c) $\frac{d^2y}{dz^2} - \frac{dy}{dz} + 5y = 0$, (d) $\frac{d^2y}{dz^2} - \frac{dy}{dz} - 5y = z$

(1) CO1 BL1

1.x) Rounding off the number 0.03709157 correct upto 5 significant figure is

- a) 0.03709
 b) 0.037091
 c) 0.037092
 d) 0.0370

(1) CO2 BL2

1.xi) If $y(x) = x^2$ and h is a step length, then find which one of the following is true

- a) $\Delta f(x) = h(x+2h)$
 b) $\Delta f(x) = 2h(x+h)$
 c) $\Delta f(x) = h(x+h)$
 d) $\Delta f(x) = h(2x+h)$

(1) CO1 BL1

1.xii) Trapezoidal rule can be applied if the number of equal sub-intervals of the interval of integration is

- a) odd
 b) even
 c) any
 d) none

(1) CO3 BL3

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. Solve $y' = 2px + yp^2$

(5) CO3 BL2

3. Apply the method of variation of parameters to solve

(5) CO3 BL3

$$\frac{d^2y}{dx^2} + 4y = \tan 2x$$

4. Using Laplace transform evaluate $\int_0^\infty t^3 e^{-t} \sin t \, dt$

(5) CO3 BL3

5. Find the missing term in the following table:

(5) CO2 BL2

x	0	1	2	3	4
f(x)	1	3	9	-	81

6. Compute $\int_{1.2}^{1.8} \left(x + \frac{1}{x} \right) dx$ using Simpson's 1/3rd rule of integration. (5) CO3 BL3
Take $n=4$.

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

- 7a. Solve by Laplace transform the equation (8) CO3 BL3
 $(D^2 + 6D + 9)y = 8 \cos t$, where $y(0) = y'(0) = 1$.
- 7b. Solve: $\frac{d^2 y}{dx^2} - 9y = e^{3x} + \cos x$ (7) CO2 BL2
- 8a. Compute y at $x=0.4$ from the differential equation $\frac{dy}{dx} = x + y^2$ with (8) CO3 BL3
initial condition $y(0.2)=1$ using Runge Kutta 4th order method. Take $h=0.1$.
- 8b. Use Lagrange Interpolation to compute $f(5)$ from the following table (7) CO3 BL3
- | | | | | |
|--------|---|----|----|----|
| x | 1 | 3 | 4 | 7 |
| $f(x)$ | 1 | 13 | 21 | 57 |
- 9a. Find, from definition, the Laplace Transform of the (7) CO2 BL2
function $f(t)$ defined by
- $$f(t) = \begin{cases} 0, & 0 < t \leq 1 \\ 1, & 1 < t \leq 2 \\ 0, & t > 2 \end{cases}$$
- 9b. Apply convolution theorem of Laplace transformation to evaluate $L^{-1} \left\{ \frac{p}{(p^2 + a^2)^2} \right\}$ (8) CO3 BL3
- 10a. Reduce the equation $\sin y \frac{dy}{dx} = \cos x (2 \cos y - \sin^2 x)$ to a linear equation (5) CO2 BL2
and hence solve it.
- 10b. Obtain the general solution and singular solution of (5) CO4 BL4
 $y = px + \sqrt{a^2 p^2 + b^2}$
- 10c. Solve: $(3x^2 y^4 + 2xy)dx + (2x^3 y^3 - x^2)dy = 0$ (5) CO2 BL2
- 11a. Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x \sin(\log x)$ (8) CO3 BL3
- 11b. Find P.I of $\frac{d^2 y}{dx^2} + 4y = x \sin x$ (4) CO3 BL2

- 11c. Show that $\Delta^r y_k = \nabla^r y_{k+r}$.

(3) CO4 BL4



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Guru Nanak Institute of Technology
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2024

EVEN SEMESTER 2023-2024
HU201 - PROFESSIONAL COMMUNICATION

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

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Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) Which of the following is an example of non verbal communication? (1) CO1 BL1

- a) Speaking
- b) Writing an email
- c) Nodding
- d) Sending a memo

1.ii) How can individuals improve their communication across cultures? (1) CO2 BL1

- a) By assuming everyone from a particular culture behaves the same
- b) By learning about cultural differences and adapting communication accordingly
- c) By avoiding interaction with people from different cultural backgrounds
- d) By imposing their own cultural norms on others

1.iii) Which type of business letter aims to persuade potential customers to purchase a product or service? (1) CO5 BL1

- a) Adjustment Letter
- b) Complaint Letter
- c) Sales Letter
- d) Job application Letter

1.iv) Which of the following is NOT a characteristic of a good report? (1) CO3 BL1

- a) Clarity
- b) Accuracy
- c) Opinions instead of facts
- d) Conciseness

1.v) When speaking on the phone, what type of communication is being used? (1) CO1 BL2

- a) Verbal and tone of voice
- b) Verbal and body language
- c) Verbal
- d) Nonverbal communication

- 1.vi) Denotations and Connotations are ——— barriers (1) CO2 BL1
in communication process.
- Physical barriers
 - Semantic barriers
 - Encoding barriers
 - Technical barrier
- 1.vii) What is the primary purpose of a CV (Curriculum Vitae) or résumé? (1) CO5 BL1
- To provide a detailed summary of personal interests
 - To highlight relevant skills, qualifications, and work experience
 - To list every job position ever held, regardless of relevance
 - To include personal opinions and beliefs
- 1.viii) Which of the following is characteristic of a chronological resume? (1) CO2 BL2
- Appropriate for experienced candidates
 - Mentions most recent job or qualification first
 - Appropriate when education and experience are unrelated to the job applied for
 - Not commonly used
- 1.ix) What is the purpose of preparing minutes of a meeting? (1) CO5 BL1
- To distribute copies of presentations and handouts
 - To summarize the discussions and decisions made during the meeting
 - To include personal anecdotes and unrelated stories
 - To showcase the creativity of the minute-taker
- 1.x) In a technical report, where would you typically find a summary of the key findings and recommendations? (1) CO4 BL1
- Conclusion
 - Abstract
 - Discussion
 - References
- 1.xi) Which of the following is NOT typically included in the minutes of a meeting? (1) CO3 BL2
- List of attendees
 - Time and date of the meeting
 - Detailed personal conversations
 - Action items and deadlines
- 1.xii) Which of the following is NOT a component of verbal communication? (1) CO1 BL1
- Tone
 - Words
 - Body language
 - Volume

Group B
(Short Answer Type Questions)

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- (Answer any three of the following) 3x5=15
2. Answer the following questions:
- Define communication. (5)
 - Communication process is incomplete without feedback. Elaborate. (2) CO1 BL1
 - Contrast between high-context and low-context cultures. (3) CO1 BL1
 - What are the similarities and differences between CVs and Resumes? Discuss briefly. (5) CO2 BL4
 - Explain the importance of including visual aids, such as graphs and charts, in a technical report. Provide examples of situations where these aids would be beneficial. (5) CO3 BL4
 - Compose an email to your professor requesting an extension for submitting a project. Ensure your email is concise, polite, and clearly explains the reason for your request. Include appropriate subject line and sign-off. (5) CO4 BL5
- Group C**
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45
7. Jupiter Infotech Services is looking for a Trainee Software Programmer with BE or B. Tech degree, conversant with programming languages, fluent in English, customer friendly and can adapt to a multicultural work environment. Write an application to the HR Manager applying for the job with your CV. (15) CO3 BL3
8. Read the passage given below and answer the questions that follow: (15)
- The culture of nuclear families is in fashion. Parents are often heard complaining about the difficulties in bringing up children these days. Too much of freedom in demand, too much independence, over night parties, excessive extravagance, splurging pocket money, no time for studies and family all this is a common cry of such families. Aren't parents, themselves, responsible for this pitiful state? The basic need of a growing youth is the family, love, attention and bonding along with moral values. One should not forget that 'charity begins at home'.
- Independence and individuality both need to be respected, in order to maintain the sanctity of family. Children, today are to be handled with tact in order to bridge the ever widening generation gap. Only the reasonable demands need to be fulfilled, as there are too many expenses to be met and too many social obligations to be taken care of by the parents. Our forefathers lived happily in joint families. Children loved to live with their cousins, learnt to adjust with them. There was perfect harmony between the generations. There never existed the concept of old-age homes. There was deep respect for the family elders and love, care and concern for the youngsters. Even the minor family differences were solved amicably.

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- a) Mention any two major common concerns of a nuclear family. Why are they problematic? (3) CO2 BL3
- b) Who, according to the passage, is responsible for these concerns? What should they be doing? (3) CO2 BL3
- c) Explain the meaning of the phrase "charity begins at home". Why is it appropriate in the present context. (3) CO2 BL3
- d) Describe the atmosphere in joint families. (3) CO2 BL3
- e) Give a suitable title to the passage. Does the writer express nostalgia for earlier times? Explain. (3) CO2 BL3
9. As the Sales Manager write a call for a meeting Notice mentioning Agenda ; also write the Minutes of a Meeting (MOM) for the launch of the Maxpro Phone 15i. Invent all necessary details. (7+8) (15) CO4 BL6
10. Discuss the role of cultural differences in workplace communication, considering the diverse cultural backgrounds of your company's employees. Suggest strategies for overcoming cultural barriers and promoting cross-cultural communication in a remote work setting. (7+8= 15) (15) CO5 BL5
11. How does nonverbal communication contribute to the overall effectiveness of interpersonal interactions, and what are some key strategies for interpreting and leveraging nonverbal cues in various social, cultural, and professional contexts? (15) CO2 BL4

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Guru Nanak Institute of Technology
An Autonomous Institute under MAKAUT
2024

EVEN SEMESTER 2023-2024
HU203 - CONSTITUTION OF INDIA

TIME ALLOTTED: 3Hours

FULL MARKS: 70

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

- 1.i) The judges of the Supreme Court can be removed by (1) CO1 BL1
- a) Chief Justice of India
 - b) President on request of Parliament
 - c) President
 - d) Prime Minister of India
- 1.ii) What is the preamble of the Indian Constitution? (1) CO1 BL1
- a) An introduction
 - b) The beginning
 - c) The preface
 - d) The prologue
- 1.iii) Under the Govt. of India Act 1935, the Indian Federation worked (1) CO1 BL2
- through which kind of list
- a) Federal List
 - b) Provincial List
 - c) Concurrent List
 - d) All the above
- 1.iv) What does Article 21 of the Indian Constitution guarantee? (1) CO1 BL2
- a) Right to Life and Personal Liberty
 - b) Right to Equality
 - c) Right against Exploitation
 - d) Right to Freedom of Religion
- 1.v) For the first time, British Parliament enacted which laws for the (1) CO1 BL2
- British Government to govern in India
- a) Govt. of India Act 1857
 - b) Govt. of India Act 1858
 - c) Govt. of India Act 1891
 - d) Govt. of India Act 1888

1. vi) Who is known as the "Father of the Indian Constitution"? (1) CO1 BL1
 a) Mahatma Gandhi
 b) Jawaharlal Nehru
 c) Dr. B.R. Ambedkar
 d) Sardar Patel

1. vii) Morely-Minto reform is associated with which Act (1) CO1 BL1
 a) Indian Council Act 1909
 b) Indian Council Act 1856
 c) Indian Council Act 1908
 d) Indian Council Act 1912

1. viii) Under the British rule, how many acts came into operation (1) CO1 BL3
 a) Govt. of India Act 1858, Indian Council Act 1861
 b) Indian Council Act 1892, Indian Council Act 1809
 c) Govt. of India Act 1919, Govt. of India Act 1935, Indian Independence Act 1947
 d) All the above

1. ix) Under the British Rule when 'Dyarchy Governing System' was introduced for the first time (1) CO1 BL1
 a) Govt. of India Act 1919
 b) Govt. of India Act 1929
 c) Govt. of India Act 1925
 d) Govt. of India Act 1935

1. x) Montagu-Chelmsford report is associated with which Govt of India Act (1) CO1 BL1
 a) Govt. of India Act 1919
 b) Govt. of India Act 1918
 c) Govt. of India Act 1920
 d) Govt. of India Act 1930

1. xi) How many Schedules are contained in the Constitution of India? (1) CO1 BL1
 a) 12 Schedules
 b) 14 Schedules
 c) 10 Schedules
 d) 15 Schedules

1. xii) The Indian Federal system is largely based on the pattern of (1) CO1 BL2
 a) US Federal system
 b) Australian Federal System
 c) Swiss Federal System
 d) Canadian Federal System

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. Write a note on the cultural and educational right of minorities in India. (5) CO3 BL2

3. Discuss some important features of Indian Independence Act of 1947 (5) CO3 BL3

4. What are the principles of Directive Principle of State Policy (DPSP)? How many articles are there under DPSP? (5) CO1 BL1

5. What are the fundamental rights? Explain briefly. (5) CO1 BL2

6. What is Lok Sabha? Give a brief idea about its function. (5) CO2 BL2

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

7. Explain the writ jurisdiction of the High court. (15) CO3 BL4

8. Discuss about the jurisdiction and function of the Supreme court of India (15) CO3 BL3

9. Discuss about the features of the Indian Constitution. (15) CO1 BL1

10. a) What are the qualifications of the member of the parliament? What are grounds for disqualification? Also explain disqualification from a political party on the ground of defection. (15) CO2 BL2

- b) Write a note on the power and functions of the State Governor. 10
5

11. a) What do you understand by independence of Supreme Court in India? 5 Marks (15) CO2 BL4

- b) Describe the Structure of the Court? 4 Marks

- c) How the Judges appointed in Supreme Court in India? 6 Marks



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Guru Nanak Institute of Technology
An Autonomous Institute under MAKAUT
2024

EVEN SEMESTER 2023-2024
HU202 - VALUES AND ETHICS

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) According to the Constitution of India, India is a (1) CO2 BL1

- a) Capitalist, Non-secular, Sovereign country
- b) Socialist, Secular, Sovereign, Democratic, Republic country
- c) Marxist, Monarchy, Secular country
- d) Socialist, Democratic, Republic country

1.ii) -----is the mother of all sciences (1) CO1 BL1

- a) Philosophy
- b) Psychology
- c) Chemistry
- d) None

1.iii) The purpose of Human centered technology is to (1) CO2 BL4

- a) Make human resources comfortable with technology
- b) Employ human technologies only
- c) Replace the big machinery with human resources
- d) All of the above

1.iv) Which writer was a major proponent of the theory of utilitarianism? (1) CO2 BL1

- a) Jeremy Bentham
- b) Immanuel Kant
- c) Karl Marx
- d) John Rawls

1.v) B.A.R.T stands for (1) CO5 BL1

- a) Bay Area Rapid Transportation
- b) Bay Area Rapid Transit
- c) Bay Area Racial Transit
- d) None

- 1.vi) What is a key characteristic of ethical leadership? (1) CO3 BL5
- Authoritarian decision-making
 - Delegating responsibility without accountability
 - Transparency and honesty in communication
 - Micromanagement of employees
- 1.vii) Which method assesses a candidate's job-related skills and abilities through practical tasks? (1) CO3 BL3
- Personality test
 - Skills test
 - Case study
 - Aptitude test
- 1.viii) How is Sustainability defined? (1) CO3 BL4
- Meeting the needs of future generation without compromising the present
 - Maximizing profits in the present without considering future needs
 - Achieving social equality in the present without considering environmental impact
 - Balancing economic growth and social welfare in the present
- 1.ix) The act of Whistle-blowing is appropriate when (1) CO4 BL2
- It is morally justified
 - It is supported and substantiated by evidences
 - All internal channels are exhausted
 - All of the above
- 1.x) Which international agreement aims to control and reduce greenhouse gas emissions? (1) CO2 BL2
- Montreal Protocol
 - Kyoto Protocol
 - Paris Agreement
 - Stockholm Convention
- 1.xi) What is meant by CSR? (1) CO5 BL1
- Corporate Social Responsibility
 - Company Social Responsibility
 - Corporate Society Responsibility
 - Company Society Responsibility
- 1.xii) Which of the following is a primary goal of pollution control measures? (1) CO5 BL2
- Maximizing industrial production.
 - Minimizing the cost of waste disposal.
 - Protecting human health and the environment.
 - Promoting economic growth without regard to environmental imp

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. Why is conducting a "social audit" advantageous for organizations? (5) CO2 BL2
3. What are the ethical issues involved in engineering profession? (5) CO3 BL3

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4. Explain the importance of Technology assessment impact analysis. (5) CO5 BL2
5. What is Whistle-blowing? Discuss the potential risks to Whistle-blowers. Discuss the cases when Whistle-blowing is appropriate and right? (5) CO4 BL4
6. Write briefly about how empathy can reduce value crisis? (5) CO1 BL2
- Group C**
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45
7. Answer the following: (15)
- Recall the founding year of the Club of Rome and identify its primary focus. (5) CO4 BL1
 - Explain the central message conveyed in "No Limits to Learning" published in 1979 and its significance in the context of education and its role in shaping a more sustainable future. (10) CO4 BL1
 - 8a. State the role of CEO in shaping business culture. (10) CO4 BL2
 - 8b. Briefly comment on the criticism of Utilitarianism. (5) CO2 BL5
 9. How can corporate environmental ethics be integrated into business practices to promote sustainable development? (15) CO5 BL3
 10. Answer the following: (15)
 - 10a. Job discrimination is a persistent issue that involves unfair treatment or prejudice against individuals based on certain characteristics – Discuss with suitable examples. (7) CO2 BL1
 - 10b. Hiring employees involves a comprehensive process of sourcing, screening, selecting, and onboarding candidates who align with the company's culture, values, and job requirements – Discuss in details. (8) CO2 BL1
 11. What do you mean by value spectrum of a good life? Briefly explain the seven human values. (8+7) (15) CO1 BL3

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